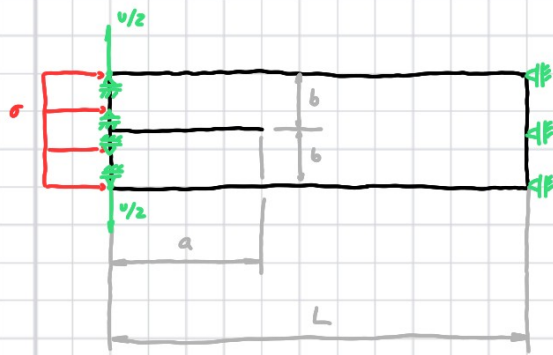


Eigenerosion



$$\begin{aligned} L &= 6 \\ b &= 0,5 \\ a &= 0,1 \\ \sigma &= 0,035 \end{aligned}$$

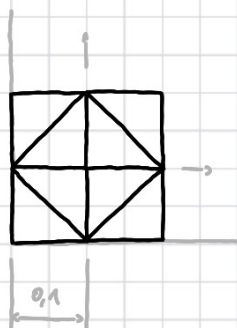
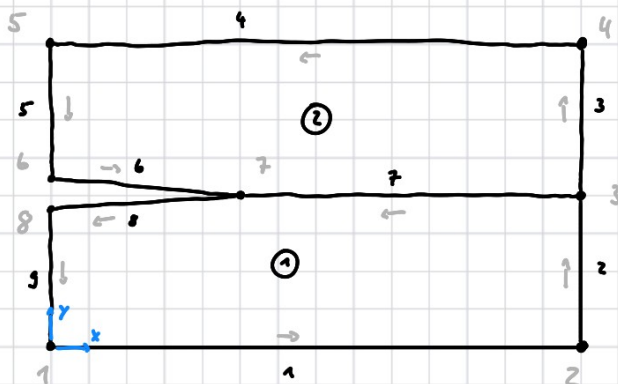
$$\begin{aligned} E &= 1,06 \\ \nu &= 0,27 \\ G_c &= 0,0001 \end{aligned}$$

$$r = \frac{3E}{G_c(1-\nu^2)} b^3$$

$$\epsilon = c_n \cdot h \quad c_n = 4$$

$$a = \frac{1}{2} \sqrt{\frac{3b^2 E r^{\frac{1}{2}} \nu}{3b^2 E + Nr^{\frac{1}{2}}(1-\nu^2)}} \quad N = \sigma b$$

Mesh



Algorithm

load step n



solve $u_h \leftarrow$



for K in cells C

K cracked?

yes

$$\varepsilon_v^K = \nabla \cdot \underline{u} < 0?$$

yes

$$\Delta E_K = \frac{1}{2} \int_{\mathcal{R}_{el}} \underline{\sigma}(\underline{u}) : \underline{\varepsilon}(\underline{u}) \, dx$$

$$\Delta A_K = \frac{1}{2\varepsilon} \sum V_K$$

$$\Delta F_K = \Delta E_K - G_c \Delta A_K$$

$$\Delta F_K > 0 \rightarrow \text{push } K \text{ in } PQ$$

if $PQ \neq \text{empty}$

$$K^* = \max(\Delta F_K \text{ from } PQ)$$

add K^* to C_ε

else

Increase loadstep

ΔA_K :



$$C^* = (C_\varepsilon \cup K) \setminus C_\varepsilon$$



$$N_\varepsilon(K) = \{j : \|x_j - x_K\| \leq \varepsilon\}$$

$$\hookrightarrow V_K$$